

Naive intervals after a recorded, effect-blind search over emulated protocols

A simulation study answering the recorded-search component of catalog problem PRO-02

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Abstract

Background. Emulation protocols are often revised after investigators see the data, and the reported interval ignores that search. Catalog entry PRO-02 states that this invalidates coverage. We measured how much coverage a naive interval loses when the search is recorded and the selector never sees the treatment effect.

Methods. Thirty-six generated-data cells crossed cohort size, expected event count, library size (4 or 16 candidate protocols over age thresholds and outcome horizons) and treatment effect, with 2,000 replicates each. Effect-blind selectors chose a protocol by arm sizes and pooled events, by prognostic associations, or by covariate balance; an effect-seeking selector was the positive control. Estimands were the global coverage deficits D_{all} and $D_{\text{nonboundary}}$ against a registered materiality threshold of 0.02. This is a registered replication: the first run failed its calibration gate by chance.

Results. All 360 fixed-candidate evaluations passed the calibration gate. D_{all} was 0.0015 (simultaneous 95% interval -0.0001 to 0.0030) and $D_{\text{nonboundary}}$ 0.0023 (0.0004 to 0.0042). The effective library size was at most 1.85. Noncoverage rose to 0.083 when candidate estimates spanned more than two standard errors. The effect-seeking control lowered coverage to 0.894 in the worst cell.

Conclusions. In this design a recorded, effect-blind search costs naive intervals almost nothing on average, because the candidates barely differ. When they do differ, the selected interval undercovers.

1 Introduction

Target trial emulations specify eligibility, treatment strategies, follow-up and outcomes before analysis, and reporting guidance asks that the specification and its revisions be recorded (1). In practice thresholds and horizons are revised after inspecting the data. Even a selector that never sees the effect estimate can use outcome frequency or covariate-outcome associations, and those are correlated with the estimate. The interval reported for the chosen protocol conditions on a choice that was itself data dependent.

PRO-02 treats this as a coverage problem. This study measures its size for the part that can be simulated: a finite, recorded search whose selector is blind to the treatment effect. Undocumented, sequential development by an analyst who has seen results is outside its scope.

2 Methods

The study follows the ADEMP structure (2). The design was reviewed and revised before any code was written.

2.1 Data-generating mechanism

Age is $55 + 10Z$ with Z a standard normal truncated at ± 2 ; S and C are binary covariates and B continuous. Baseline initiation depends on Z , S , C and B , and time to event follows a constant-hazard model with the same covariates. The treatment effect is null, a hazard ratio of 0.75, or a hazard ratio $\exp(-0.288 + 0.25Z)$ that changes with age, so candidate protocols with different age thresholds target different effects. The baseline hazard was solved, before the study and from event counts only, so that the anchor protocol (age at least 55, 36 months) has 35, 70 or 140 expected pooled events. There is no loss to follow-up. The four-candidate library crosses age thresholds of 50 and 55 with horizons of 36 and 60 months; the sixteen-candidate library crosses thresholds of 45 to 60 with horizons of 24 to 60 months.

2.2 Estimands and selectors

For each candidate the estimand is the risk difference at its horizon among people meeting its age threshold. Each protocol's effect was estimated by Hajek inverse probability weighting with a correctly specified propensity model and a Wald interval. Selectors: a workability rule (the first candidate with at least 100 per arm and 70 pooled events); two prognostic-association rules, whose fixed score tracks the outcome more or less closely; an outcome-blinded rule that picks the eligibility threshold with the best covariate balance; and, as a positive control, the candidate with the most beneficial standardized estimate. Two remedies were evaluated: Bonferroni over the declared library and a fifty-fifty split between a design half that selects and an analysis half that estimates.

2.3 Decision rule

A calibration gate comes first: every declared candidate in every cell must have fixed-candidate coverage compatible with $[0.93, 0.97]$, judged by a Wilson interval at familywise confidence $1 - 0.05/360$, and failure probability compatible with at most 0.05. If it passes, the problem is materially real if both simultaneous 95% lower limits of D_{all} and $D_{\text{nonboundary}}$ exceed 0.02, not materially real if both upper limits are below 0.02, and mixed otherwise.

2.4 Why this is a replication

The first run used the same design with a gate that required all 360 coverage estimates to lie inside $[0.93, 0.97]$. One came in at 0.9285, 0.26 Monte Carlo standard errors outside, and the registered classification was held at uninformative. A gate applied without allowance for Monte Carlo error to 360 estimates fails almost surely even when every candidate is calibrated. The replication protocol, committed before any of its replicates were drawn, changed only the seed and the gate. The first run's deficits, 0.0003 and 0.0000, were also far below the threshold.

3 Results

3.1 Calibration and the primary comparison

All 360 fixed-candidate evaluations passed the gate; fixed-candidate coverage ran from 0.9295 to 0.9610 and no candidate failed to estimate. D_{all} was 0.0015 (-0.0001 to 0.0030) and $D_{\text{nonboundary}}$ 0.0023 (0.0004 to 0.0042). Both upper limits lie more than 0.015 below the threshold. The registered branch is that the recorded-search problem is not materially real in this design. The nonboundary deficit is detectably above zero but equals about a fifth of a percentage point.

3.2 Why the average deficit is small

Nested age thresholds and horizons produce highly correlated candidates. The effective library size (the squared sum of the eigenvalues of the candidates' correlation matrix divided by the sum of their squares) was 1.36 to 1.40 with 4 candidates and 1.75 to 1.85 with 16. Choosing among them is close to not choosing.

3.3 The design can detect a loss

The effect-seeking selector lowered naive coverage to 0.930 on average and to 0.894 to 0.928 across the 16-candidate cells, against 0.947 to 0.949 for the effect-blind selectors (Table 1).

Table 1: Mean naive coverage of the selected protocol’s interval, by selector and treatment effect.

selector	heterogeneous	homogeneous	null
effect_seeking	0.934	0.936	0.922
outcome_blinded	0.950	0.948	0.950
prognostic_high	0.951	0.948	0.949
prognostic_low	0.948	0.947	0.946
workability	0.950	0.948	0.949

3.4 The average hides a conditional failure

Naive noncoverage depended on how far the candidates’ estimates spread (Table 2). When the range across candidates was below one median standard error, noncoverage was 0.022 to 0.025; between one and two, 0.041 to 0.044; above two, 0.082 to 0.083. The standardized range predicted noncoverage with an AUC of 0.66 to 0.68.

Table 2: Naive noncoverage by the standardized range of the candidate estimates (range divided by the median candidate standard error).

selector	below 1	1 to 2	above 2
prognostic_high	0.022	0.041	0.083
prognostic_low	0.025	0.044	0.083
workability	0.023	0.041	0.082

3.5 Remedies

Bonferroni intervals for the selected protocol covered 0.9915, above nominal. Sample splitting covered 0.9468, with intervals a median 1.42 times as wide as the full-sample interval for the same candidate, close to the square root of two.

4 Discussion

A recorded search whose selector cannot see the effect does little damage to average coverage when the candidates are variations on one protocol. The damage concentrates in the replicates where the candidates disagree, which is also where the choice matters. Reporting every candidate’s estimate, as a multiverse display, therefore carries information about the reliability of the selected interval that the interval itself does not.

Bonferroni and sample splitting both restore coverage. Bonferroni overshoots; splitting pays in width about what halving the data predicts.

4.1 What this study does not establish

The result covers deterministic recorded searches over nested age thresholds and horizons, libraries of 4 or 16 highly correlated candidates, effect-blind selectors and correctly specified weights. It does not cover sequential protocol development by an analyst who has seen results, other protocol components such as grace periods or treatment definitions, libraries of weakly correlated protocols, or misspecified weights. The replication’s gate controls false failure across 360 comparisons and would pass a candidate miscalibrated by less than about two percentage points.

4.2 Review

The design was reviewed and revised before any code was written. The first run was reviewed twice; the reviewer held its classification at uninformative and recommended a separately registered replication, which is this study. The replication’s results have not yet been reviewed.

Data and code availability

Code, both protocols and result files: <https://github.com/choxos/TTE-open-problems/tree/main/studies/PRO-02-ben-02-recorded-protocol-search>.

References

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